



温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W02
Strategic Mgt Global Info Sys

Strategic Use of Information Resources

Day 2 · Afternoon Session

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Learning Objectives

By the end of this session, you should be able to:

- 1 Understand what is information resource**
IT assets + IT capability
- 2 Apply three analytical lenses**
Five Forces, Value Chain, Resource-Based View
- 3 Distinguish operational, temporary, and sustained advantage**
Information resource → short-lived advantage, or durable value

MORNING — inside the firm

The IS Strategy Triangle

- Business, organizational, and information strategies must pull together.
- Alignment is what makes working technology deliver value.
- Misalignment is the most common reason IS investments under-perform.

Key question: does IS support the firm's strategy?

AFTERNOON — looking outward, inward, and deeper

Strategic use of information resources

- Is the firm's competitive environment putting the right pressure on its IS?
- Where inside the firm does IS actually add value?
- Will that value last — or will competitors copy it away?

Key question: does IS also give the firm an advantage competitors cannot copy?

Three Lenses

A single map we will refer back to throughout the afternoon.

LENS 1

Five Forces

Outside the firm

Where is the pressure?

Identifies the competitive forces shaping the industry and shows how MIS can shift them in the firm's favor.

LENS 2

Value Chain

Inside the firm

Where does MIS add value?

Maps the primary and support activities where MIS improves efficiency, coordination, or customer experience.

LENS 3

Resource-Based View

Sustaining the firm

Does the advantage last?

Asks whether the MIS capability is valuable, rare, and hard enough to imitate to create sustainable advantage.

Running Case — Zara

The Zara business model

- Flagship brand of Inditex, headquartered in Arteixo, Spain.
- ~12,000 new designs per year; ~450 million items manufactured; production runs usually under 8,000 pieces.
- Concept-to-store: ~2 weeks for modifications, 4–5 weeks for new designs — vs. 3–6 months industry norm.
- Regular customers visit ~17 times per year — industry average is ~4.
- More than 50% of product manufactured proximally (Spain, Portugal, Turkey, Morocco).
- Strategy summarized by two rules: "Give customers what they want" and "Get it to them faster than anyone else."

12,000

new designs per year

2 weeks

concept-to-store for modifications

17 visits

per regular customer per year

Information Resource

Information resources come in two flavors — and only one of them can be bought.

IT ASSETS

Anything — tangible or intangible — that a firm can use to create, produce, or offer products.

IT infrastructure

- Platforms — OS, mobile, cloud
- Company apps
- Network and data
- Website

Information repository

- Customer and employee information
- Marketplace and vendor information
- Platform-partner information

IT CAPABILITIES

Skills and know-how developed over time — what makes IT assets actually work for the firm.

Technical skills

Systems analysis, development, data science, network design.

IT management skills

Business-process knowledge, vendor evaluation, project management.

Relationship skills

- Spanning — connecting IS with internal business units
- External — managing vendors, platform partners, customers

Assets can be purchased. Capabilities are built over time — and that is why they matter more.

The Eras Model — Six Decades of Information Usage

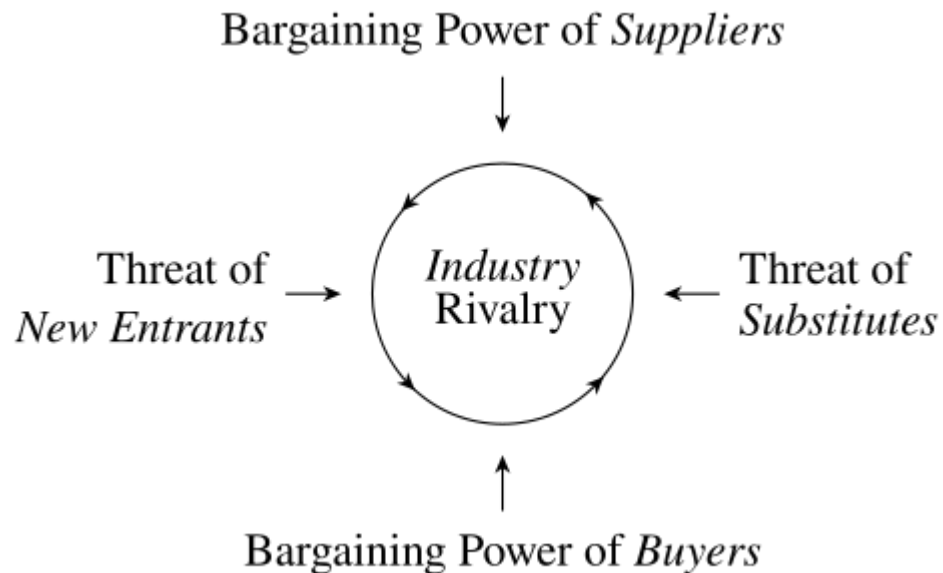
Understanding how information resources have evolved helps us see what is genuinely new about today's strategic use.

Era I	Era II	Era III	Era IV	Era V	Era VI+
1960s	1970s	1980s	1990s	2000s	2010s→
Role of IT Efficiency ----- Value from Scarcity	Role of IT Effectiveness ----- Value from Scarcity	Role of IT Strategy ----- Value from Scarcity	Role of IT Strategy ----- Value from Plenitude	Role of IT Value creation ----- Value from Plenitude	Role of IT Value from ecosystems ----- Value from Hyper-plenitude

The shift around Era IV

Network effects broke the old logic: value no longer scales with scarcity but with plenitude. An email account is more valuable the more people have one. This is why cloud, social, analytics, and AI reshape competitive dynamics — not because the tech itself is powerful, but because abundance creates network value.

Lens 1 – Porter's Five Forces



[Source: Wikipedia](#)



[Michael Porter](#)

- A framework for analyzing a firm's **competitive environment** via five forces.
- *Can MIS make a force less punishing?*

MIS Through Five Forces Lens



Force	How MIS Can Help
Industry rivalry	Differentiates or lowers cost structurally
Threat of new entrants	Raises barriers to entry
Threat of substitutes	Reduces the price-performance gap
Bargaining power of buyer	Increase switching costs / personalization
Bargaining power of suppliers	Create substitution options / tighter integration

Zara Through the Five Forces Lens

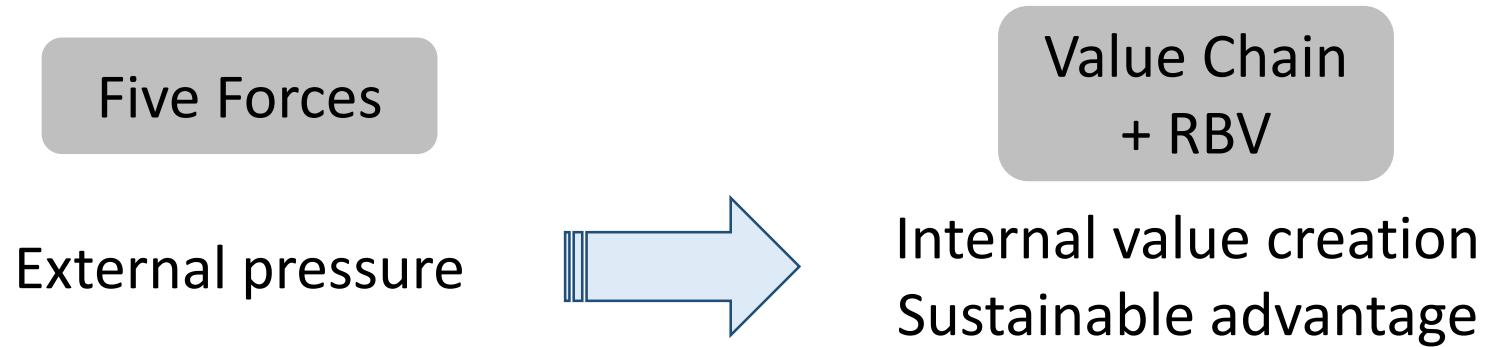
Analysis

- Fashion retail is highly competitive; speed and trend responsiveness matter
- Retail entry is possible, but fast-response execution is harder to build
- Customers have many alternatives; speed and design reduce substitute pressure
- Information sharing and coordination reduce supplier dependence

Discussion

- 1) Which force is **strongest**?
- 2) How does **MIS** help Zara respond?
- 3) Is MIS creating advantage, or just helping Zara keep up?

From Pressure to Value Creation



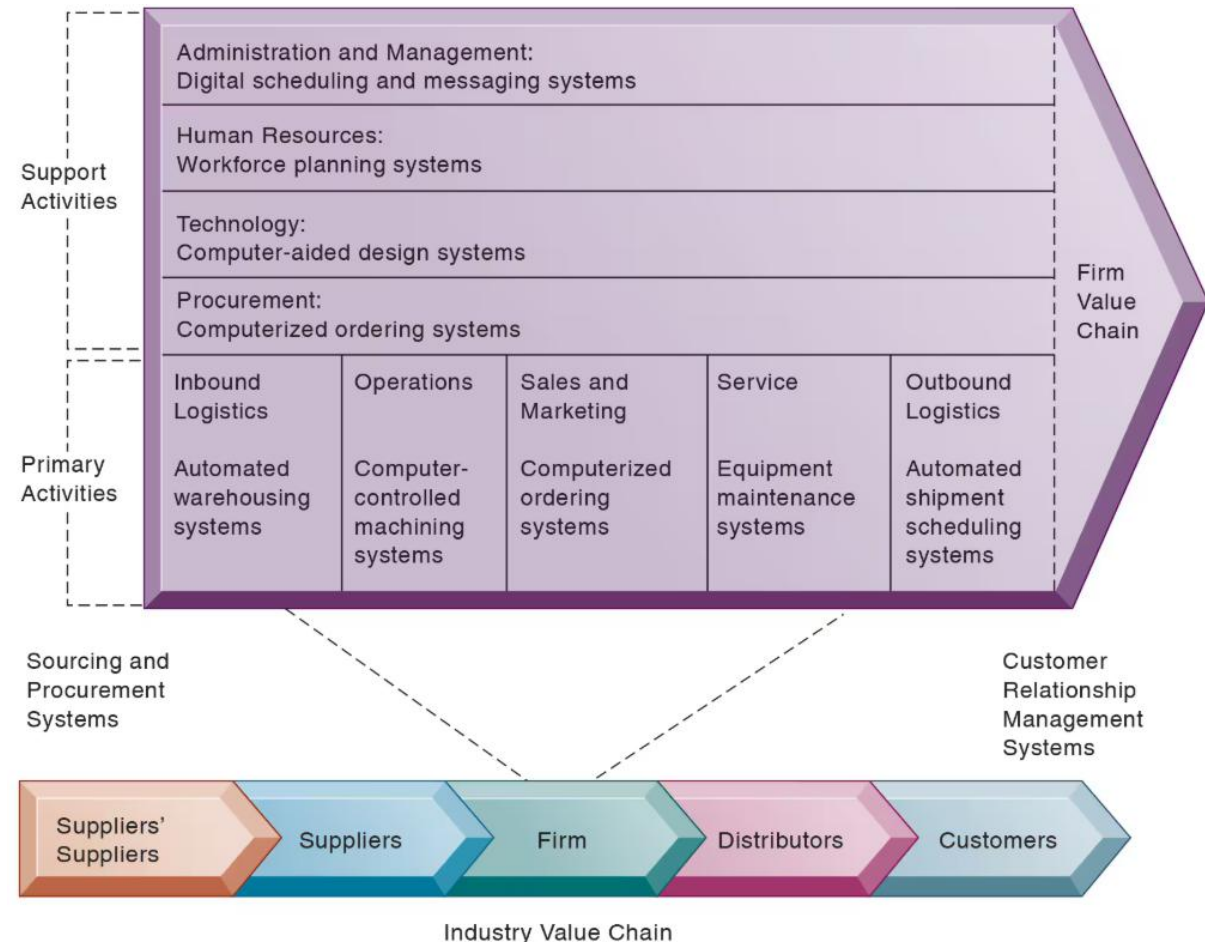
Key Questions

- Where does MIS add value inside the firm?
- When does that value become hard to copy?

Lens 2 – Porter's Value Chain

A sequence of activities that an organization performs to design, produce, market, deliver, and support goods or services for customers.

[Source: Wikipedia](#)

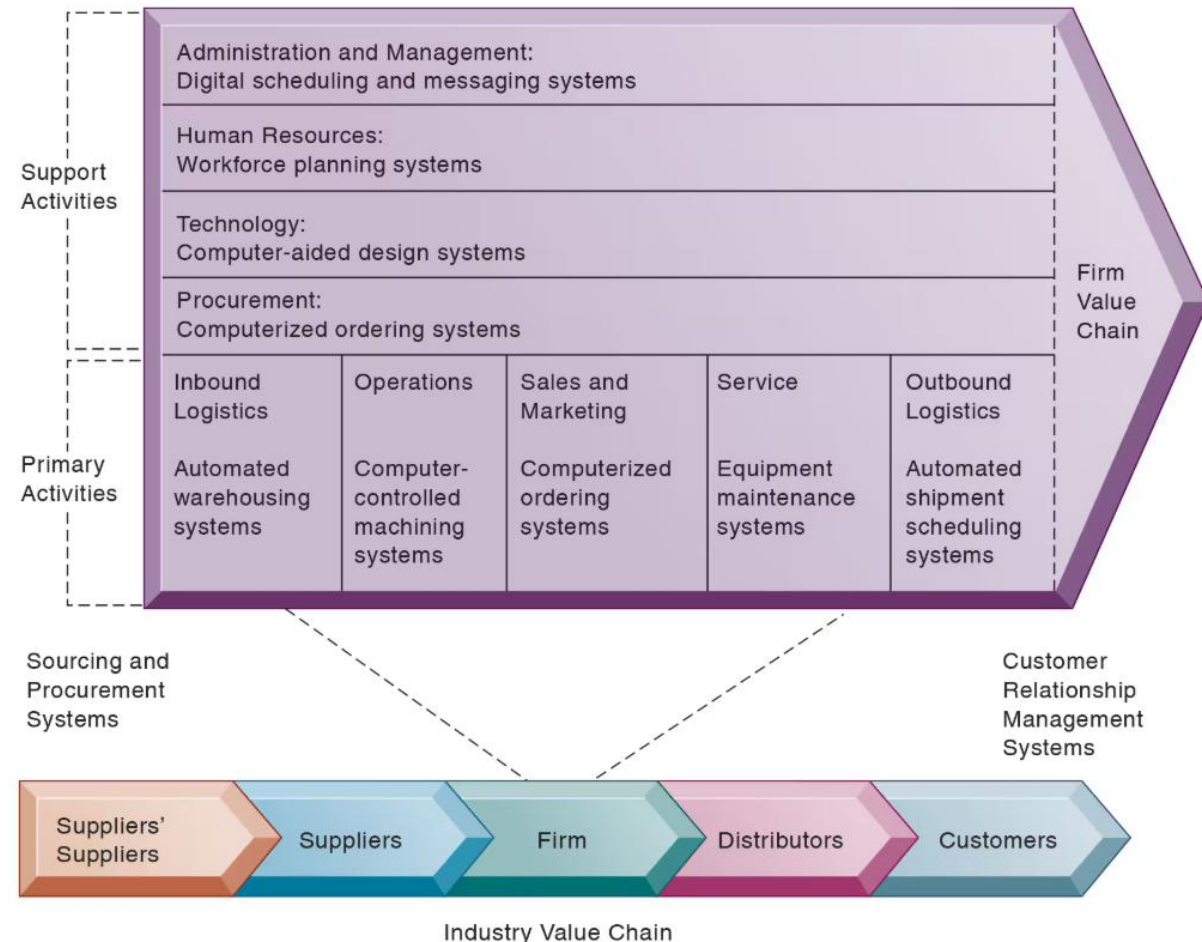


Lens 2 – Porter's Value Chain

Primary activities directly create customer value through core processes:

→ Inbound logistics, operations, outbound logistics, sales & marketing, and service.

(From receiving materials → producing goods → delivering → selling → maintaining products.)

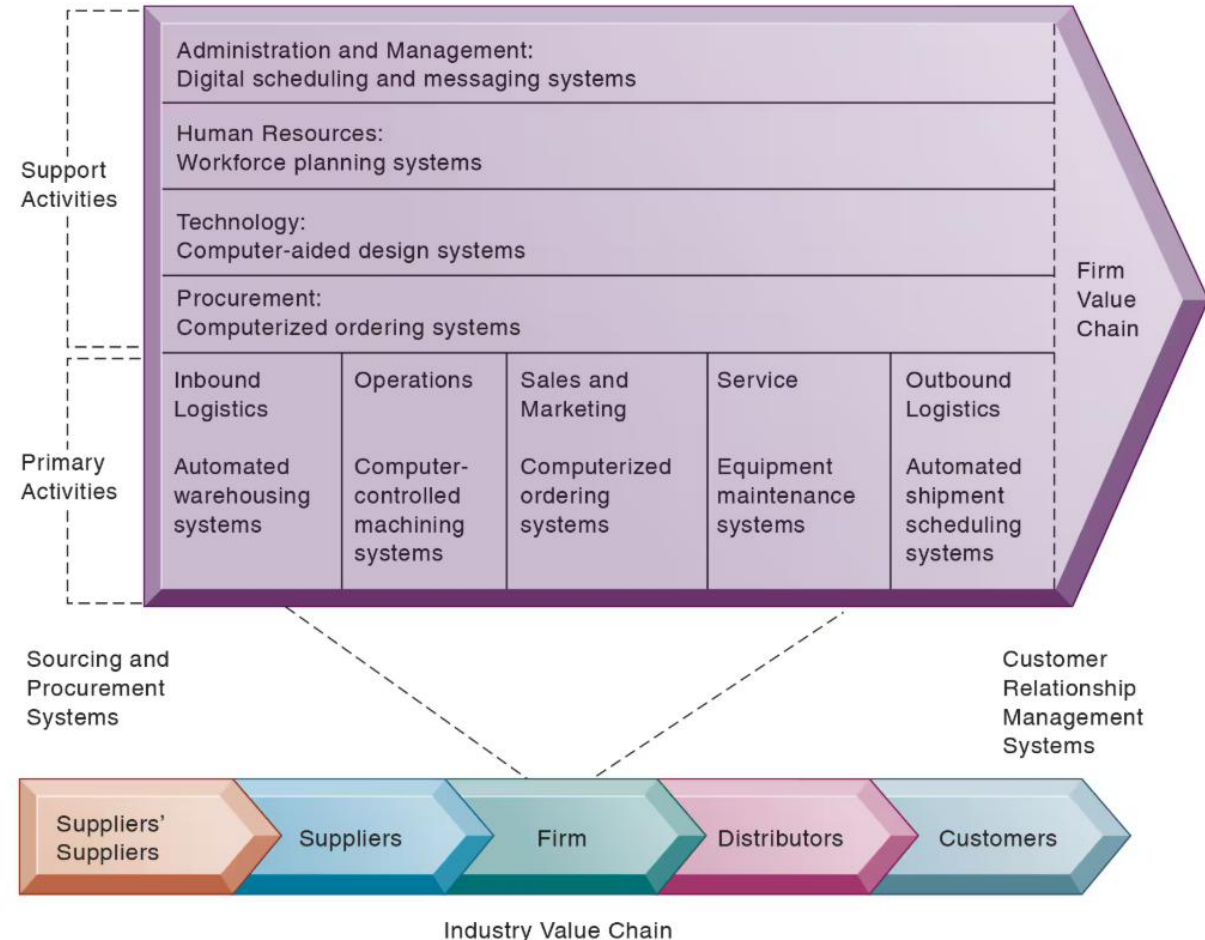


Lens 2 – Porter's Value Chain

Support activities enable and enhance primary activities through:

→ Infrastructure, human resources, technology development, and procurement.

(They provide the foundation that allows core operations to run efficiently.)



Zara Through the Value Chain

	Activity	How MIS shows up at Zara
Primary	Operation	12 highly automated factories; robots cut and dye; computer-controlled equipment positions patterns across up to 100 layers.
	Outbound logistics	Miles of automated conveyor belts sort and consolidate shipments. Store-level replenishment arrives within ~48 hours.
	Marketing & sales	POS at each shop floor records every sale; store managers submit twice-weekly orders and daily notes via a proprietary smartphone app.
Support	Technology	In-house IT team works directly with conveyor vendors to co-develop distribution technology.
	Human resources	Tight coupling across designers, store managers, market specialists, and production planners — enabled by shared platforms.

RBV: Why It Matters for MIS

Owning technology is easy. **Building advantage from it is harder.**

RBV helps us ask:

- Is this information resource valuable?
- Is it rare?
- Is it difficult to imitate?
- Is it difficult to substitute?
- Is the firm organized to use it well?

A standard system rarely creates lasting advantage by itself.

Lens 3 – Resource-Based View (VRIO)

Core idea

Competitive advantage comes from a firm's information resources when those resources are both valuable and hard for competitors to copy. Owning technology is easy; building advantage from it is hard.

To Attain Advantage

- **Valuable:** *makes the firm more efficient, effective, or innovative*
- **Rare:** *not already held by most rivals — otherwise it is just "table stakes"*

To Sustain Advantage

- **Hard to imitate:** *embedded in routines, relationships, culture*
- **Hard to substitute:** *no equivalent that delivers the same value*
- **Hard to transfer:** *cannot simply be purchased or hired away*

What Is Easy to Copy – and What Is Not?

Easier to Copy

- Standard ERP
- Common cloud tools
- Basic dashboards
- Off-the-shelf analytics

Harder to Copy

- Proprietary data
- Cross-functional routines
- Strong IT-business collaboration
- Fast decision loops
- Culture of using data in operations

The MIS advantage is often not the system itself, but how the organization uses it.

Sustainability Barriers

Barrier	Description
IT project barrier	The initiative is difficult, costly, or time-consuming to replicate
IT assets and capabilities barrier	The firm has data, skills, or integration capability others lack
Complementary resources barrier	Advantage depends on brand, process, supplier network, or culture in addition to IT
Preemption barrier	The firm moves first and locks in customers, partners, or data

Piccoli, G., & Ives, B. (2005). IT-dependent strategic initiatives and sustained competitive advantage: A review and synthesis of the literature. MIS Quarterly, 747-776.

How Innovation Sustain Advantage

Even the strongest advantages erode.
Sustainability requires continuous reinvention — not stability.

1. Enhance Value with IT

- Improve products, services, and customer experience
- Use data and AI to continuously refine offerings

2. Anticipate Disruption

- Act before competitors or new entrants
- Compete with your future self, not just rivals

3. Balance Horizons

- Short-term: incremental improvements
- Long-term: business model transformation

Example: Netflix

- DVD rental → Streaming platform → Content producer
- Proactively cannibalized its own model
- Accepted short-term losses to lead industry shift

→ Turned risk into long-term dominance

Zara Through the RBV Lens

Questions

- Is Zara's MIS valuable?
- Is it rare?
- What exactly is hard to imitate?

Focus

- Not just software
- Fast data-to-decision process
- Tight coordination among stores, design, and operations
- Embedded routines and organizational discipline

The real MIS advantage often comes from integration and use, not ownership alone.

Zara Through the RBV Lens

<i>Resource/Attribute</i>	<i>Value Creation</i>		<i>Value Sustainability</i>		
	<i>Value</i>	<i>Rarity</i>	<i>Imitation</i>	<i>Substitution</i>	<i>Transfer</i>
IT Infrastructure	Moderate – effective use of POS, handhelds, and automation, but technologies are replicable by competitors		Easy to imitate and transfer its infrastructure Moderate for substitution of infrastructure (automated conveyers)		
Information Repository	High – valuable and rare customer data, tightly integrated with operations and design decisions		Difficult to imitate and transfer Extremely difficult to substitute because of the volume and nature of the data		

Pearlson et al. 2024

Zara Through the RBV Lens

Resource/Attribute	Value Creation		Value Sustainability		
	Value	Rarity	Imitation	Substitution	Transfer
Technical Skills	Low — limited value/rarity; technical talent is widely available		Moderately difficult to imitate or substitute; offers some sustainability through cross-system integration		
IT Management Skills	High — valuable and rare, accumulated over time		Difficult to imitate, substitute, or transfer; resources leveraged well		
Relationship Skills — <i>Externally Focused</i>	High value — strong supplier relationships Moderate rarity — competitors can also built relationships with suppliers		Difficult to imitate, substitute, or transfer — ~5-week end-to-end turnaround		
Relationship Skills — <i>Spanning</i>	High rarity of spanning		Difficult to imitate, substitute, or transfer — tight-knit teams enable fast, accurate responses to customer needs		

Dynamic Capabilities — RBV for turbulent markets

Teece's dynamic capabilities view focuses on how firms create, extend, and reconfigure their resources in response to change.

Sensing

WHAT IT IS

Spotting opportunities and threats in the environment before competitors do.

AT ZARA

Daily sales reports from every store to designers. Store managers trained to listen to what customers want but could not find.

Seizing

WHAT IT IS

Mobilizing resources to act on what was sensed — through investment, commitment, and speed.

AT ZARA

Agile collaborative teams across designers, buyers, and operations act on latest trends within days.

Transforming

WHAT IT IS

Reconfiguring the firm's resources, structures, and culture as the environment shifts.

AT ZARA

Proximal manufacturing plants get latest trends to stores within 48 hours; the entire supply chain is built for agility.

Sources: Teece, Pisano & Shuen (1997); Pearlson et al. (2024), Figure 2.8

Managerial Takeaway

Advantage from information resources has three layers — and they are not equally durable.

1

Operational support

Systems that keep the firm running. Necessary — but rivals can buy the same thing. Treat as table stakes.

2

Temporary advantage

A valuable and rare capability that has not yet been copied. Enjoy the lead — and invest in the next one before imitation catches up.

3

Sustained advantage

Capabilities embedded in routines, data, and relationships that competitors cannot easily copy, substitute, or transfer. This is where strategic IS pays back.

Synthesis

Lens 1 – Five Forces

Where is the pressure?

Identifies the competitive forces shaping the industry and shows how MIS can shift them in the firm's favor

How MIS responds
to competition

Lens 2 – Value Chain

Where does MIS add value?

Maps the activities where MIS improves efficiency, coordination, or customer experience inside the firm.

Where MIS
adds value

Lens 3 – RBV

Does the advantage last?

Asks whether the MIS capability is valuable, rare, and hard enough to imitate to create sustainable advantage.

Whether MIS creates
sustainable advantage

Does this MIS merely support operations, or does it change how the firm competes?

Synthesis Case – Zara

How this run

Groups of four or five. 15 minutes of structured discussion, then 20 minutes of plenary. Each group prepares the answer to ONE assigned question and summarizes it in 90 seconds.

The five questions are designed to INTEGRATE lenses — no single lens answers any question alone.

Five Forces	<i>Where is the pressure Zara has neutralized — and where not?</i>
Value Chain	<i>Which activity is IS most indispensable to?</i>
RBV / VRIO	<i>Which information resources sustain Zara's advantage, and which are table stakes?</i>
Dynamic capabilities	<i>How do sensing, seizing, and transforming play out at Zara?</i>
Managerial decision	<i>What is the next IS investment Zara should make?</i>

Connecting today to April 25 and to your project

Next Saturday — Session 3

MIS and Digital Transformation Ethical and Social Issues of MIS

- From using IS to compete, to redefining the firm through digital.
- When strategy and IS fuse — digital business strategy.
- Case: Tsingtao Brewery's digital transformation.

Group Project

Stage 1 Proposal

- Apply all three lenses to your chosen firm's MIS.
- Classify each information resource as asset vs. capability.
- Map at least one primary activity and one support activity where MIS is material.
- **Proposal due May 9 (Sat), 23:59**

Reference and AI Use Disclosure

Primary readings

Pearlson, K. E., Saunders, C. S., & Galletta, D. F. (2024). Strategic use of information resources. In *Managing and Using Information Systems: A Strategic Approach* (8th ed., Chapter 2). Wiley.

Laudon, K. C., Laudon, J. P., & Traver, C. G. (2025). Information systems, organization and strategy. In *Management Information Systems: Managing the Digital Firm* (18th ed., Chapter 3). Pearson.

Case sources

Zara material adapted from Pearlson et al. (2024), Chapter 2 opening vignette and Figures 2.3, 2.4, 2.5, 2.7, 2.8.

Complementary sources

Porter (1985); Porter & Millar (1985); Barney (1991); Teece, Pisano & Shuen (1997); Piccoli & Ives (2005); Wade & Hulland (2004).

AI use disclosure

AI was used as a thinking and design assistant to:

- Redesign the session flow around three integrated lenses.
- Collapse redundant navigation slides into a single canonical map.
- Build a running Zara case that supports all three lenses.
- Develop discussion prompts that force integration of lenses.

Instructor role

All content was reviewed, selected, and adapted by the instructor for this course context.